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Frontend Assignment

Ans 1) CSS Box Model

The CSS Box Model is a fundamental concept that describes how every HTML element is displayed as a rectangular box on a web page. Every element is made up of four parts:

1. Content - the actual text, image or data inside the element.
2. Padding - space between the content and the border.
3. Border - a line surrounding the content and padding.
4. Margin - space outside the border that separates the element from other elements.

The total size of an element is calculated by adding up the content, padding, border and margin. If `box-sizing: border-box` is used, the width and height already include the padding and border, which makes sizing more predictable.

Ans 2) Media Query in CSS

A media query is used to make a website responsive. It allows different styles to be applied based on screen size, device type or resolution, without changing the HTML.

Example:

```
body {  
  background-color: white;  
}  
  
@media (max-width: 600px) {  
  body {  
    background-color: lightblue;  
  }  
}
```

In this example the background colour is white by default, but changes to light blue when the screen width becomes 600px or smaller.

Ans 3) Important Features of Bootstrap

Bootstrap is a popular front-end framework used for creating responsive websites quickly.

Features:

- Responsive grid system
- Predefined CSS classes
- Ready-made components such as buttons, cards, forms and navigation bars
- Built-in JavaScript components

- Mobile-first design approach
- Easy customization
- Faster web development

Bootstrap reduces development time and keeps the design consistent across all pages of website.

Ans 4) Bootstrap Grid System

Bootstrap uses a 12-column grid system to create responsive layouts.

Example:

```
<div class="container">
  <div class="row">
    <div class="col-4">Column 1</div>
    <div class="col-4">Column 2</div>
    <div class="col-4">Column 3</div>
  </div>
</div>
```

Explanation:

- container creates the main layout area.
- row creates a horizontal group of columns.
- col-4 occupies 4 out of 12 columns, so the three col-4 divs together occupy all 12 columns and sit side by side.

Ans 5) CSS Selectors

CSS selectors are used to target HTML elements and apply styles to them.

a) Element selector

```
p {
  color: red;
}
```

Styles all paragraph elements.

b) Class selector

```
.box {
  color: blue;
}
```

Styles all elements having class "box".

c) ID selector

```
#heading {
  color: green;
}
```

Styles the element with id "heading".

d) Universal selector

```
* {
```

```
margin: 0;
}
```

Applies the style to all elements.

e) Group selector

```
h1, p {
  color: black;
}
```

Applies the same style to multiple elements.

Ans 6) CSS Background Properties

Background properties are used to control the appearance of an element's background. Important properties are:

- background-color
- background-image
- background-repeat
- background-position
- background-size
- background-attachment
- background (shorthand for all of the above)

These properties help improve the visual appearance of a webpage.

Ans 7) Overlapping Elements in CSS

Elements can be overlapped using the position and z-index properties.

Example:

```
#box1 {
  position: absolute;
  left: 50px;
  top: 50px;
  z-index: 1;
}
```

```
#box2 {
  position: absolute;
  left: 80px;
  top: 80px;
  z-index: 2;
}
```

The element with the higher z-index value (box2) appears above the element with the lower z-index value (box1). Note that z-index only works on elements that are positioned, i.e. position: static.

Ans 8) CSS Position Property

The position property specifies how an element is placed on a web page.

Types of position:

1. Static

Default position of an element. It follows the normal document flow.

```
position: static;
```

2. Relative

Moves the element relative to its own original position, while still keeping the space originally occupied.

```
position: relative;
```

```
top: 20px;
```

3. Absolute

Positioned relative to the nearest positioned ancestor. If there is no such ancestor, positioned relative to the page itself.

```
position: absolute;
```

```
top: 20px;
```

4. Fixed

Positioned relative to the browser window, so it remains fixed in place even when the page is scrolled.

```
position: fixed;
```

```
bottom: 10px;
```

5. Sticky

Acts like relative until a certain scroll position is reached, after which it sticks in place like fixed.

```
position: sticky;
```

```
top: 0;
```

Ans 9) ECMAScript

ECMAScript is the standard specification on which JavaScript is based. It defines the syntax, features and behaviour of the JavaScript language.

ECMAScript includes:

- Variables
- Data types
- Functions
- Objects
- Classes
- Loops
- Modern JavaScript features

Example:

ES6 (ECMAScript 2015) introduced several important features such as let, const, arrow functions, classes and template literals. These features make JavaScript more powerful and easier to use.

Ans 10) Difference between var, let and const

Feature	var	let	const
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Scope	Function scope	Block scope	Block scope
Re-declaration	Allowed	Not allowed	Not allowed
Reassignment	Allowed	Allowed	Not allowed
Hoisting	Yes	Yes	Yes
Temporal Dead Zone	No	Yes	Yes

Temporal Dead Zone (TDZ)

The Temporal Dead Zone is the time period between entering a block and the point where a var or const variable is declared. During this period, the variable exists but cannot be accessed.

Example:

```
console.log(x);  
let x = 10;
```

This code produces an error because x is accessed before its declaration.